Lecture 1: Foundations and Key Concepts

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Today's Journey

Part 1: Foundations

- What is Machine Learning?
- A Motivating Example
- Key Components of ML

Part 2: Problem Types

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

Part 3: Deep Learning

- The Road to Deep Learning
- Why Deep Learning Works
- Success Stories

Learning Outcomes

- Define ML and its components
- Distinguish problem types
- Understand deep learning's role

The Fundamental Problem

Traditional Programming

Human writes rules + Data $\longrightarrow$ Computer $\longrightarrow$ Output

Machine Learning Approach

Data + Desired Outputs $\longrightarrow$ Computer $\longrightarrow$ **Rules (Model)**

Key Insight: The computer discovers the rules itself!

What is Machine Learning?

Definition

A computer program is said to learn from **experience E** With respect to some class of **tasks T** and **performance measure P**, If its performance at tasks in T as measured by P improves with experience E.

Contrast with Traditional Programming:

- Traditional: follows the same business logic forever
- ML: improves automatically with more data/experience

Deep Learning: A powerful subset of ML techniques driving innovations in computer vision, NLP, healthcare, and genomics.

A Motivating Example: Wake Word Recognition

The Challenge: Write a program to recognize "Hey Siri" or "Alexa"

Why It's Hard:

- 44,000 audio samples per second
- Each sample = amplitude measurement
- Write function f(raw audio) $\rightarrow$ {yes, no}?

The ML Solution:

- Collect labeled audio snippets
- Define a **function** with **parameters**
- Use data to find best parameters

The Trick

Even if we don't know how to *program* wake word recognition, we can *perform* it ourselves $\rightarrow$ we can collect labeled examples!

Programming with Data

The Training Process

1. Start with a **randomly initialized model**
2. Grab some data (e.g., audio + labels)
3. **Tweak parameters** to improve performance
4. Repeat until the model is awesome!

Key Concepts:

1. **Parameters** “Knobs” that control program behavior
2. **Model** The program with fixed parameters
3. **Model Family** All possible programs from parameter choices
4. **Learning Algorithm** The “meta-program” that chooses parameters

Key Components of Machine Learning

1. Data

- Collection of **examples**
- Each has **features** (inputs)
- Supervised: has **labels**
- More data $\rightarrow$ better models
- "Garbage in, garbage out"

2. Model

- Transforms data $\rightarrow$ predictions
- Deep learning: many successive transformations

3. Objective Function

- Measures performance
- Also called **loss function**
- Convention: lower is better
- E.g., Squared error, Cross-entropy

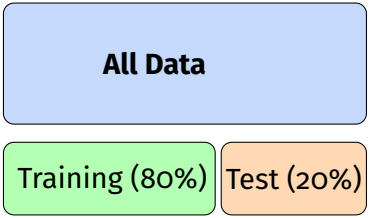
4. Optimization

- Searches for best parameters
- **Gradient descent**: adjust parameters to lower loss

Training vs. Test Data

Why Split the Data?

- **Training set**: used to learn parameters
- **Test set**: held out for evaluation
- Goal: generalize to *unseen* data



Overfitting

When a model performs well on training data but fails to generalize to unseen data.

Analogy: A student memorizes practice exams but fails the real test.

Supervised Learning

Definition

Tasks where we have a dataset containing both **features** and **labels**, and we produce a model that predicts labels given new features.

The supervision: We provide labeled examples during training.

Examples:

- Predict cancer vs. not cancer from CT images
- Predict French translation from English sentence
- Predict stock price from financial data

Supervised learning accounts for the majority of successful ML applications in industry.

Supervised Learning: Regression

Definition

Predict **continuous numerical values**.

Example: House Price Prediction

- Features: square footage, bedrooms, bathrooms, location
- Label: sale price (a number)

Rule of Thumb: Any "*how much?*" or "*how many?*" problem is likely regression.

More Examples:

- How many hours will this surgery take?
- How much rainfall will this town have?
- What rating will a user give this movie?

Common Loss Function: Squared error (assumes Gaussian noise)

Supervised Learning: Classification

Definition

Predict which **category** (class) an example belongs to.

Binary Classification (2 classes)

- Spam / Not Spam
- Disease / Healthy
- Fraud / Legitimate

Multiclass Classification

- Digit recognition: 0, 1, 2, ..., 9
- Character recognition: a, b, c, ...

Probabilistic Output

- Model outputs probability per class
- Example: 90% cat, 10% dog
- Conveys uncertainty

Common Loss Function:
Cross-entropy

Key: Most likely class $\neq$ always best decision (consider costs!)

Other Supervised Learning Tasks

Tagging (Multi-label)

Predict multiple classes.

- Blog tags: “ML”, “Code”
- Image: cat AND dog

Recommender Systems

Personalized suggestions.

- Netflix movies
- Amazon products
- Implicit feedback

Search & Ranking

Rank items by relevance.

- Web search results
- PageRank algorithm

Sequence Learning

Variable-length inputs.

- Machine translation
- Speech recognition
- Text-to-speech

Unsupervised Learning

Definition

Learning patterns from data **without labels**.

"The boss hands you a giant dump of data and tells you to do some data science with it!"

Key Questions:

- **Clustering:** Can we group photos into landscapes, dogs, cats, mountains?
- **Dimensionality Reduction:** Can we find a small number of parameters that capture the data? (e.g., Principal Component Analysis)
- **Representation Learning:** Can we represent objects so "Rome" − "Italy" + "France" = "Paris"?
- **Distribution Learning:** What is the probability distribution of these data?
- **Generative Models:** Can we generate new realistic data?

Reinforcement Learning

Definition

An **agent** interacts with an **environment**, takes **actions**, and receives **rewards**.

The RL Loop:

1. Agent observes environment state
2. Agent chooses an action
3. Environment gives reward
4. Agent updates its **policy**
5. Repeat

Key Challenges:

- **Credit Assignment:** Which actions led to reward?
- **Partial Observability:** Can't see full state of environment
- **Explore vs. Exploit:** Try new things or stick with what works?

- **Goal:** Learn a policy that maximizes cumulative reward.
- **Famous Examples:** AlphaGo, Atari games, robotics, autonomous driving

Comparing ML Paradigms

Aspect	Supervised	Unsupervised	Reinforcement
Labels	Yes	No	Rewards
Feedback	Direct	None	Delayed
Goal	Predict labels	Find patterns	Maximize reward
Data	Static dataset	Static dataset	Interactive
Examples	Classification	Clustering	Game playing
	Regression	PCA	Robotics

Note: These paradigms can be combined! RL can use supervised pre-training; unsupervised features can improve supervised learning.

The Road to Deep Learning

What Changed?

Data Explosion

- World Wide Web
- Millions of online users
- Cheap sensors
- Inexpensive storage

Compute Revolution

- GPUs (originally for gaming)
- Moore's Law
- Parallel processing
- Cloud computing

Decade	Dataset Size	Memory	Compute
1990	10K samples	10 MB	10 MF
2000	10M samples	100 MB	1 GF
2010	10B samples	1 GB	1 TF
2020	1T samples	100 GB	1 PF

Key Advances in Deep Learning

Architectures

- **Dropout** - capacity control
- **Attention mechanisms** - learnable pointers
- **Transformers** - superior scaling
- **GANs** - generative modeling
- **Diffusion models** - DALL-E, Imagen

Infrastructure

- **Parallel training** - 1000s of GPUs
- **Frameworks** - PyTorch, TensorFlow, JAX
- **Pre-trained models** - transfer learning
- **Large Language Models** - ChatGPT, GPT-4

The Transformer Revolution

Built solely on attention, Transformers excel across NLP, computer vision, speech, reinforcement learning, and more. A single model can play Atari, caption images, chat, AND control robots!

The Essence of Deep Learning

What Makes Deep Learning "Deep"?

Models learn **many layers of transformations** - each layer learns representations jointly from data.

Key Principle: End-to-End Training

- **Traditional:** Feature engineering $\rightarrow$ then train model
- **Deep Learning:** Learn features AND model together

What Deep Learning Replaced:

- Hand-crafted features (Canny edge detector, SIFT)
- Shallow models at the end of pipelines
- Domain-specific preprocessing

Result: Unified tools for computer vision, NLP, speech, medical informatics, and more!

Deep Learning Success Stories

Perception

- **Speech Recognition:** Human parity achieved
- **Image Recognition:** Error rate 28% $\rightarrow$ 2.25%
- **Intelligent Assistants:** Siri, Alexa, Google Assistant

Games

- DeepBlue (Chess)
- AlphaGo (Go)
- Libratus (Poker)
- Atari games

Applications

- Self-driving vehicles
- Medical diagnosis
- Fraud detection
- Search & recommendations
- Machine translation

Creative AI

- DALL-E, Midjourney
- ChatGPT
- Text-to-speech
- Music generation

Challenges and Considerations

Data Quality Issues

- Garbage in, garbage out
- Underrepresented groups
- Historical biases in data
- Feedback loops in recommendations

Societal Impact

- Job automation
- Algorithmic bias
- Privacy concerns
- Need for procedural fairness

The Bigger Picture

"It is important to ensure that these algorithms are used with care. Statistical models, when applied without care, can lead to racial, gender, or age bias."

Summary

1. **Machine Learning** = algorithms that learn from experience
2. **Key Components:** Data, Model, Objective Function, Optimization
3. **Three Paradigms:**
   - Supervised (labels) $\rightarrow$ Classification, Regression
   - Unsupervised (no labels) $\rightarrow$ Clustering, Generation
   - Reinforcement (rewards) $\rightarrow$ Sequential decisions
4. **Deep Learning:** Multi-layer representations learned end-to-end
5. **Enablers:** Big data, GPUs, better algorithms, open frameworks
6. **Remember:** Quality data matters; consider societal impact

Exercises for Reflection

1. Which parts of code you currently write could be “learned”? What data would you need?
2. What problems have many examples but no obvious automation approach? (Prime candidates for deep learning!)
3. How do data characteristics and computational resources influence algorithm choice?
4. Where might end-to-end training be useful but isn’t currently the default approach?

Next Chapter: Preliminaries - Linear Algebra, Calculus, Probability

Thank You!

Questions?

Resources:

`https://d2l.ai`

Interactive notebooks available for PyTorch, TensorFlow, JAX

"The best way to understand deep learning is learning by doing."